

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1
library(tidyverse)
library(agricolae)
library(ggplot2)
library(here)
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
ntl.lter <- read.csv(
  here("Data", "Raw", "NTL-LTER_Lake_ChemistryPhysics_Raw.csv"),
  stringsAsFactors = TRUE)

ntl.lter$sampldate <- as.Date(ntl.lter$sampldate, format = "%m/%d/%Y")

#2 Set theme
```

```
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "right")
theme_set(mytheme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

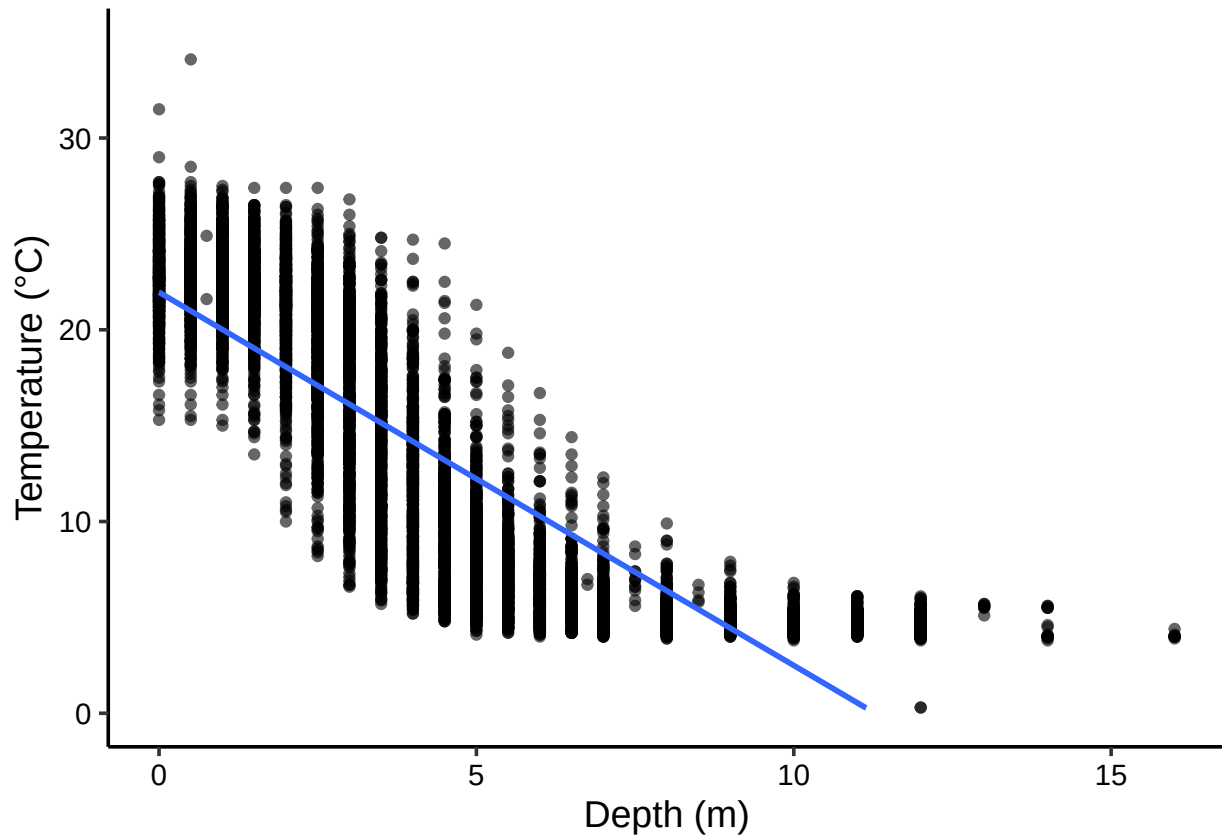
3. State the null and alternative hypotheses for this question: > Answer: H0: The mean lake temperature recorded during July does not change with depth. Ha: Across all lakes, the mean lake temperature recorded during July changes with depth.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

#4 Data Wrangling

```
ntl.lter.selected.july <- ntl.lter %>%
  filter(month(sampledate) == 7) %>%
  select(lakename, year4, daynum, depth, temperature_C) %>%
  drop_na()
```

#5 Visualize the relationship between depth and temperature

```
ggplot(ntl.lter.selected.july, aes(x = depth, y = temperature_C)) +
  geom_point(alpha = 0.6) + # scatter points
  geom_smooth(method = "lm", se = TRUE) + # linear model
  scale_y_continuous(limits = c(0, 35), name = "Temperature (°C)") +
  scale_x_continuous(name = "Depth (m)")
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: In general, the linear trend line suggests a negative relationship between the depth and the temperature. In other words, as the depth increases, the temperature decreases, which is consistent with the overall distribution of the points. However, it is worth noting that the majority of the points are clustered around the left-hand side of the plot and the trend starts to flatten at greater depths, which could indicate a nonlinear trend as the lake temperature stabilizes at bottom layers.

7. Perform a linear regression to test the relationship and display the results.

#7 Linear Regression

```
lm.temp.depth <- lm(temperature_C ~ depth, data = ntl.lter.selected.july)
summary(lm.temp.depth)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth, data = ntl.lter.selected.july)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 21.95597    0.06792   323.3  <2e-16 ***
## depth      -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The linear regression shows a strong, statistically significant negative relationship between the two variables. The model indicates that for every 1-meter increase in depth, the temperature decreases by around 1.946°C, which is statistically significant as the p-value is extremely small ($< 2.2e-16$).

The model explains about 73.87% of the total variation in temperature ($R^2 = 0.7387$), based on 9726 degrees of freedom. The high R^2 value and degree of freedom suggest that the depth is a strong indicator of temperature variation.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

#9 AIC to determine the variables

```
temp.regression <- lm(data = ntl.lter.selected.july, temperature_C ~
                      year4 + daynum + depth)
step(temp.regression)
```

```
## Start: AIC=26065.53
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141687 26066
## - year4      1         101 141788 26070
## - daynum     1         1237 142924 26148
## - depth      1      404475 546161 39189
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = ntl.lter.selected.july)
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##    -8.57556      0.01134      0.03978     -1.94644
```

#10 According to AIC, the best fit is to include all the variables

```
temp.regression.best <- lm(data = ntl.lter.selected.july, temperature_C ~
                           year4 + daynum + depth)
summary(temp.regression.best)
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = ntl.lter.selected.july)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The multiple regression model is: `temperature_C ~ year4 + daynum + depth` The three explanatory variables together provide the best model fit according to AIC.

The model explains about 74.1% of the variation in temperature. It is a strong fit and a small improvement from the depth-only model. First, the R^2 value increases, which means the new model explains more variance now. Second, in the AIC method, the multiple regression model has a lower AIC value. It also makes sense in real-life scenarios to include `year4` and `daynum` as variables to capture the seasonal and long-term variations.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
#12 ANOVA model
```

```
# Expressed as an ANOVA model
```

```
temp.lake.anova <- aov(data = ntl.lter.selected.july,
                      temperature_C ~ lakename)
summary(temp.lake.anova)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Expressed as a linear model
```

```
temp.lake.anova2 <- lm(data = ntl.lter.selected.july,
                      temperature_C ~ lakename)
summary(temp.lake.anova2)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = ntl.lter.selected.july)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake       -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake      -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake    -6.5972     0.6769  -9.746 < 2e-16 ***
## lakenameWard Lake       -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake  -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

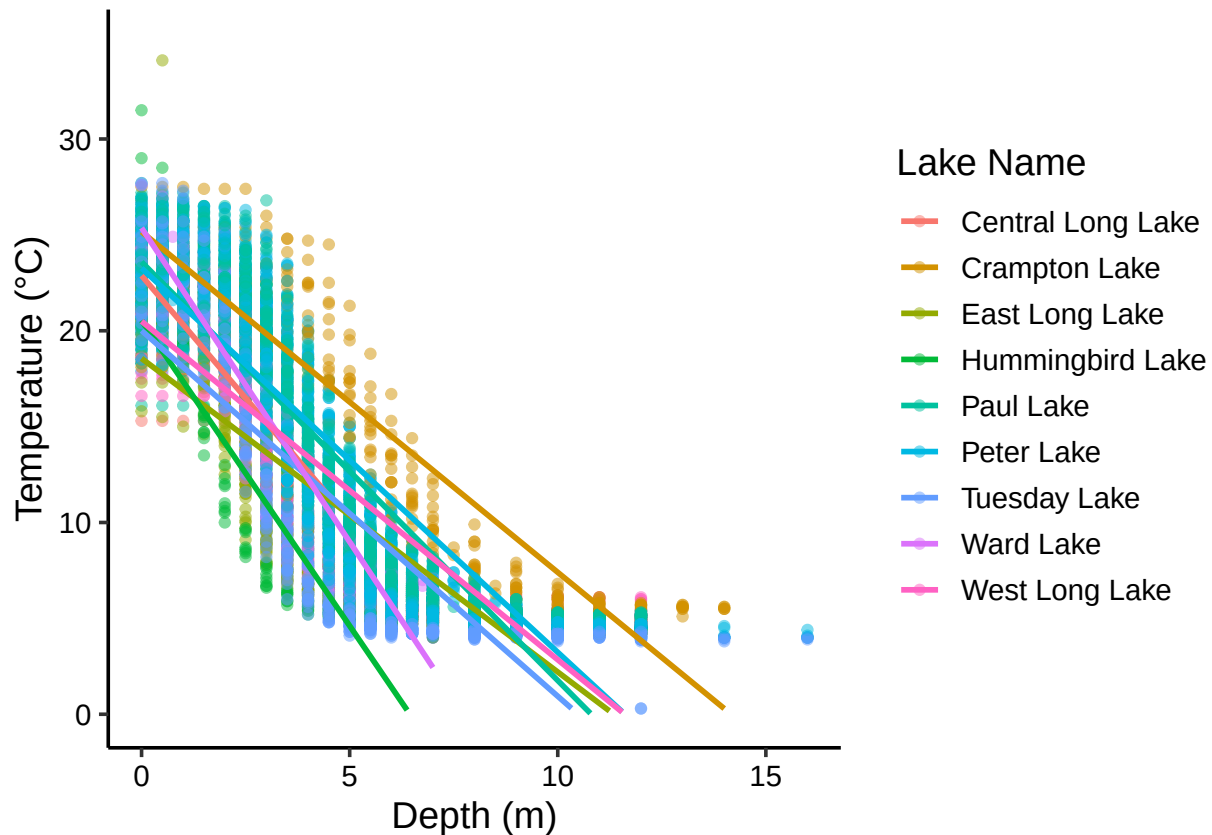
Answer: Yes - there is a significant difference among the lakes. The ANOVA test result shows a p-value $< 2e-16$, which means the mean temperature among the lakes has a significant difference. It suggests that some lakes have different mean temperatures far greater than would be expected by random variation alone.

When the test is expressed as a linear model, the overall F-test is significant, and all the individual lakes all show significant differences, with p-values < 0.01 . Meanwhile, it is worth noting that the model only explains for 3.87% of the variance, which indicates that lake name only accounts for a small fraction of variability.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

#14.

```
ggplot(ntl.lter.selected.july, aes(depth, temperature_C, color = lakename)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_y_continuous(limits = c(0, 35), name = "Temperature (°C)") +
  scale_x_continuous(name = "Depth (m)") +
  theme(legend.position = "right") +
  labs(color = "Lake Name")
```



15. Use the Tukey's HSD test to determine which lakes have different means.

```

#15
tukey.result <- TukeyHSD(temp.lake.anova)
tukey.result

## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = ntl.lter.selected.july)
##
## $lakename
##
## diff lwr upr p adj
## Crampton Lake-Central Long Lake -2.3145195 -4.7031913 0.0741524 0.0661566
## East Long Lake-Central Long Lake -7.3987410 -9.5449411 -5.2525408 0.0000000
## Hummingbird Lake-Central Long Lake -6.8931304 -9.8184178 -3.9678430 0.0000000
## Paul Lake-Central Long Lake -3.8521506 -5.9170942 -1.7872070 0.0000003
## Peter Lake-Central Long Lake -4.3501458 -6.4115874 -2.2887042 0.0000000
## Tuesday Lake-Central Long Lake -6.5971805 -8.6971605 -4.4972005 0.0000000
## Ward Lake-Central Long Lake -3.2077856 -6.1330730 -0.2824982 0.0193405
## West Long Lake-Central Long Lake -6.0877513 -8.2268550 -3.9486475 0.0000000
## East Long Lake-Crampton Lake -5.0842215 -6.5591700 -3.6092730 0.0000000
## Hummingbird Lake-Crampton Lake -4.5786109 -7.0538088 -2.1034131 0.0000004
## Paul Lake-Crampton Lake -1.5376312 -2.8916215 -0.1836408 0.0127491
## Peter Lake-Crampton Lake -2.0356263 -3.3842699 -0.6869828 0.0000999
## Tuesday Lake-Crampton Lake -4.2826611 -5.6895065 -2.8758157 0.0000000
## Ward Lake-Crampton Lake -0.8932661 -3.3684639 1.5819317 0.9714459
## West Long Lake-Crampton Lake -3.7732318 -5.2378351 -2.3086285 0.0000000
## Hummingbird Lake-East Long Lake 0.5056106 -1.7364925 2.7477137 0.9988050
## Paul Lake-East Long Lake 3.5465903 2.6900206 4.4031601 0.0000000
## Peter Lake-East Long Lake 3.0485952 2.2005025 3.8966879 0.0000000
## Tuesday Lake-East Long Lake 0.8015604 -0.1363286 1.7394495 0.1657485
## Ward Lake-East Long Lake 4.1909554 1.9488523 6.4330585 0.0000002
## West Long Lake-East Long Lake 1.3109897 0.2885003 2.3334791 0.0022805
## Paul Lake-Hummingbird Lake 3.0409798 0.8765299 5.2054296 0.0004495
## Peter Lake-Hummingbird Lake 2.5429846 0.3818755 4.7040937 0.0080666
## Tuesday Lake-Hummingbird Lake 0.2959499 -1.9019508 2.4938505 0.9999752
## Ward Lake-Hummingbird Lake 3.6853448 0.6889874 6.6817022 0.0043297
## West Long Lake-Hummingbird Lake 0.8053791 -1.4299320 3.0406903 0.9717297
## Peter Lake-Paul Lake -0.4979952 -1.1120620 0.1160717 0.2241586
## Tuesday Lake-Paul Lake -2.7450299 -3.4781416 -2.0119182 0.0000000
## Ward Lake-Paul Lake 0.6443651 -1.5200848 2.8088149 0.9916978
## West Long Lake-Paul Lake -2.2356007 -3.0742314 -1.3969699 0.0000000
## Tuesday Lake-Peter Lake -2.2470347 -2.9702236 -1.5238458 0.0000000
## Ward Lake-Peter Lake 1.1423602 -1.0187489 3.3034693 0.7827037
## West Long Lake-Peter Lake -1.7376055 -2.5675759 -0.9076350 0.0000000
## Ward Lake-Tuesday Lake 3.3893950 1.1914943 5.5872956 0.0000609
## West Long Lake-Tuesday Lake 0.5094292 -0.4121051 1.4309636 0.7374387
## West Long Lake-Ward Lake -2.8799657 -5.1152769 -0.6446546 0.0021080

# Another way to do Tukey's HSD test
#tukey.result2 <- HSD.test(temp.lake.anova, "lakename", group = TRUE)
#tukey.result2

```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter

Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Peter Lake-Paul Lake and Peter Lake-Ward Lake have the same mean temperature, statistically speaking, because their pairwise comparisons have $p > 0.05$. It suggests their means do not differ significantly.

In the given dataset, no lake has a mean temperature that is distinct from all the other lakes. For each of the lake, at least one pairwise comparison has $p > 0.05$. Some lakes do differ less than the most. For example, Central Long Lake only has one p -value > 0.05 .

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: We can use two-sample t-test, which compares the means of two independent groups to see if they are significantly different.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
# Wrangle the July data
ntl.lter.selected.july.lakes <- ntl.lter.selected.july %>%
  filter(lakename %in% c("Crampton Lake", "Ward Lake"))

# Two-sample T-test
twosample.result <- t.test(temperature_C ~ lakename,
                           data = ntl.lter.selected.july.lakes)

twosample.result
```

```
##
## Welch Two Sample t-test
##
## data: temperature_C by lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -0.6821129 2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##           15.35189                14.45862
```

Answer: The t-test result shows a p-value of 0.2649, which is larger than 0.05 and not small enough for us to reject the null hypothesis. Hence, there is no significant difference in means between Crampton Lake and Ward Lake. The 95% confidence interval that includes zero also supports the conclusion.

The mean temperature of Crampton Lake is slightly higher than the one of Ward Lake. In the Tukey's HSD test result, we also see a p -value > 0.05 , suggesting that the mean temperatures of the two lakes do not differ significantly with each other. The two results are consistent.